

| User Story                                                                                                                                                     | To-Do | In Progress | Done                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| As a student, I want to see my classes placed accurately on a map of UCSC with restricted boundaries and interactive popups.                                   |       |             | <p>Configure the Google Maps JS SDK to initialize with a custom mapId. Implement LatLngBounds to restrict the user's view and panning to the UCSC campus perimeters. (4)</p> <p>Develop the "handshake" between the Go API and the frontend that maps building names from the database to their corresponding Latitude/Longitude coordinates. (5)</p> <p>Create a robust engine that clears and redeploys markers based on search results, ensuring Lat/Lng data is converted into interactive map pins in real-time. (5)</p> <p>Engineer the "Info Box" logic that triggers on pin-clicks. This includes formatting the HTML content to display room numbers, lecture times, and instructors, and ensuring the popup is responsive and "filter-aware" (hides meetings if they are filtered out). (6)</p> |
| As a user, I want a fast and responsive interface so that I can interact with my course data in real-time, anywhere, without the entire page having to reload. |       |             | <p>Update CSS to ensure all interactive elements (search buttons, course cards, filter checkboxes) have a minimum touch target of 44x44px. (1)</p> <p>Create a CSS animation for "loading" cards that appear in the sidebar while the Go API is fetching results. (2)</p> <p>Rewrite the core layout in style.css so that mobile is the default. Use media queries to apply the 350px sidebar only on screens wider than 768px. (8)</p> <p>Update script.js so that when the sidebar is toggled on mobile, it slides over the map (z-index) rather than pushing the map container. (2)</p> <p>Wrap the search input in a &lt;form&gt; and implement event.preventDefault() to allow mobile "Enter" keys to search without refreshing the page. (1)</p>                                                    |
| As a student, I want to search and see colored pins on a building to distinguish between different classes and meeting types.                                  |       |             | <p>Refactor the clustering logic so that if multiple meeting types occur in one building, the "Group" (Circle) icon is rendered with the number of meetings in the middle. (8)</p> <p>Implement crisp SVG paths for ★ (LEC), ■ (LAB), and ▲ (DIS) within the createMarkerElement function for the Google Maps Advanced Marker API. (4)</p> <p>Add JS listeners to markers so that hovering over a pin triggers a "bounce" or "scale" animation on the corresponding sidebar card. (4)</p> <p>Audit and refactor the ColorManager to ensure the same hex code is used for the sidebar border, the map marker, and the InfoWindow header for a specific class. (2)</p> <p>Replace the current color array with a hand-picked set of 20 high-contrast colors to ensure markers are distinguishable. (1)</p>  |
| As a student, I want to add all my classes to one view so I can visualize my entire weekly commute across campus.                                              |       |             | <p>Implement map.fitBounds() logic that triggers whenever a new class is added, ensuring the zoom level fits all active pins. (3)</p> <p>Update the sidebar rendering to display primary lecture times (e.g., "MWF 10:40AM - 11:45AM") directly on the course cards for quick reference. (3)</p> <p>Ensure the searchCourse function appends new results to the global state rather than overwriting previous searches. (2)</p> <p>Create an "Add All Saved" button in the sidebar that iterates through localStorage and pushes all saved items into the active map view. (4)</p>                                                                                                                                                                                                                        |
| As a student, I would like to have full control over the visibility of each meeting type and the ability to manage my saved list.                              |       |             | <p>Add visibility control icons to individual course cards, allowing users to hide or show specific sections on the map without removing the data from their current results. (4)</p> <p>Ensure the toggleSaveCourse function maintains absolute parity between the UI state and localStorage, ensuring the "Saved for Later" list persists through page reloads. (2)</p> <p>Refine the checkbox filtering system to perform high-performance batch updates on marker visibility (LEC vs. DIS vs. LAB) across the entire map instance. (3)</p> <p>Update the InfoWindow generation engine to respect active filters; if "Labs" are globally hidden, they should automatically be omitted from the text content of the building popups. (3)</p>                                                            |